

MACHINE LEARNING ENGINEER ASSESSMENT

Date: August 2026

Freight Rate Prediction & Forecasting Project

End-to-End Supervised Learning Model, Explainable AI (SHAP), Daily December Forecast, and Production REST API Architecture.

GitHub Repositorygithub.com/pruthvirajtarode/freight-rate**Live Web Dashboard**freight-rate-one.vercel.app

Executive Performance Metrics

GradientBoo

WINNING MODEL

0.8507HOLDOUT R² SCORE**6.76%**

HOLDOUT MAPE

\$128.98

HOLDOUT MAE

1. Dataset & Data Pipeline Architecture

The system ingests 48,000 historical shipment records to predict `posted_rate` (continuous USD freight rate). Preprocessing is designed to avoid data leakage while ensuring robust feature handling.

Split / Dataset	Record Count	Strategy & Description
Training Set	40,800 rows (85%)	IQR outlier filtering (0.15% dropped), model training, 5-fold CV
Internal Holdout	7,200 rows (15%)	Unseen validation evaluate overall generalization (MAE: \$128.98)
Validation Assessment	12,000 rows	Unlabeled holdout file (<code>validation_predictions.csv</code> generated)
December Forecast	31 rows	Daily December 2025 input features for seasonal curve generation

2. 5-Fold Cross-Validation Benchmarks

Evaluated 7 distinct regression algorithms using 5-fold cross-validation on the training set. **GradientBoosting** outperformed all competitors on RMSE and R^2 .

Algorithm	5-Fold CV RMSE (\$)	CV R^2 Score	Fit Time (sec)	Status
Gradient Boosting	375.01	0.9272	9.65s	 Selected Winner
LightGBM	376.39	0.9266	0.22s	Runner-up
CatBoost	385.36	0.9230	0.50s	Evaluated
Linear Regression	387.53	0.9222	0.03s	Baseline

Algorithm	5-Fold CV RMSE (\$)	CV R ² Score	Fit Time (sec)	Status
Random Forest	390.11	0.9212	5.70s	Evaluated
Extra Trees	392.76	0.9201	3.04s	Evaluated
XGBoost	401.32	0.9166	0.50s	Evaluated

3. Model Interpretability & SHAP Feature Drivers

To ensure transparency, global SHAP (Shapley Additive exPlanations) values were calculated. The primary decision drivers for freight rates include:

- Top Feature Contributions:**
- Distance (miles):** Strongest positive driver. Rates scale non-linearly over long haul lanes.
 - Cargo Weight (lbs):** Higher equipment weight increases rate per mile.
 - Rate per Mile (Engineered):** Interaction term providing clear baseline expectations.
 - Market Capacity Index:** Captures spot market tight/loose capacity supply dynamics.

4. December 2025 Daily Forecast Insights

The model was applied to the 31 daily input records for December 2025. Key seasonal dynamics captured:

- Pre-Holiday Surge (Dec 18–22):** Peak rate spike driven by holiday shipping volume constraints.
- Christmas Dip (Dec 24–25):** Temporary rate decline due to market operational pauses.
- Year-End Adjustment (Dec 29–31):** Stabilization as annual contracts reset.

5. API Architecture & Mobile Responsive Web App

The solution includes a full single-port FastAPI backend serving both endpoints and the interactive glassmorphism Web SPA dashboard:

Endpoint Path	HTTP Method	Functionality
/api/v1/predict/single	POST	Real-time single shipment prediction with feature breakdown
/api/v1/predict/batch	POST	Bulk CSV processing & downloadable predictions
/api/v1/metrics	GET	Returns model MAE, RMSE, R ² , MAPE, and hyperparameter config
/static/charts/shap_summary.png	GET	Static file serving for explainability plots

Verification & Mobile Responsiveness

All 5 automated tests passed (pytest project/backend/tests/). The frontend uses flexbox, grid layouts, and custom media queries for full mobile responsiveness across all devices.